

The statistics of soaring flight

EconophysicsLab offsite 2026



Matteo Di Sante

Transport, routes and
launch environments

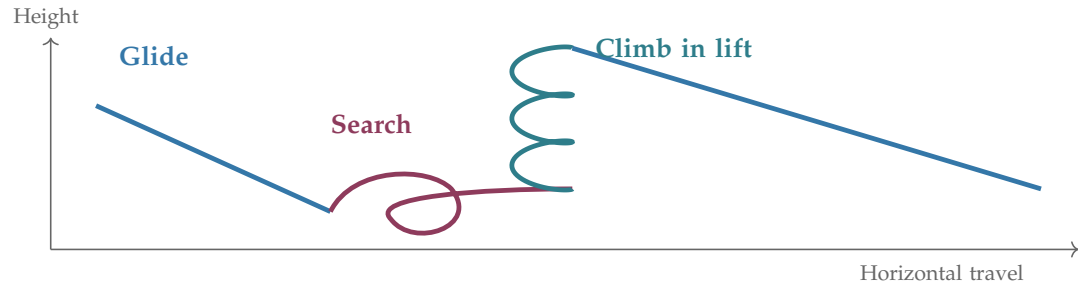
EconophysicsLab offsite 2026 | Slide 1

The statistics of soaring flight

The aim is to describe soaring flight as a transport process and to identify empirical constraints for a later stochastic model. The trajectory combines pilot decisions, the search for lift, aircraft performance and the surrounding atmosphere. Today I cover the archive and its preparation, then the horizontal mean-square displacement and the conditional comparisons through Section 3.2 of the current thesis. The line on the cover is a schematic, not a measured flight.

Soaring is transport with an energy budget

Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Schematic

A flight is sustained by an energy budget: the pilot trades altitude for horizontal distance, searches for rising air, and recovers altitude in lift. These are useful behavioural descriptions, not phase labels supplied by the recorder or results of a segmentation shown here. Ground motion includes both air-relative motion and wind. The challenge is to explain whole-flight transport while respecting these physical constraints. The present measurements are the first layer of that programme. An MSD exponent alone cannot distinguish all the candidate mechanisms: persistent motion, changes of behaviour and heterogeneous conditions can produce similar second moments. The schematic is illustrative and its horizontal axis has no numerical scale. Source: thesis introduction, Sections 1.1–1.3.

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

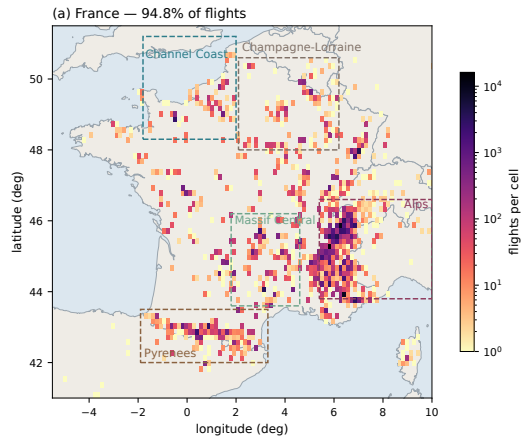
FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

The source is the FFVL Coupe Fédérale de Distance, with separate archives for paragliders and hang gliders. The verified downloads contain 186,052 and 6,716 tracklogs. Season names refer to their starting year. The IGC files supply time-stamped coordinates and altitude channels, while the catalogue supplies flight date, declared take-off site, route class and wing class. The map shows the retained population, not the raw downloads; both disciplines are pooled. About 94.8 percent of retained flights are in metropolitan France, with a strong Alpine concentration. Additional flights occur in La Réunion and elsewhere. This is a submitted cross-country archive with uneven geographic coverage, not a random sample of all free flight. Sources: thesis Chapter 2; generated/stats.tex, pipeline_census.tex and prelim_map.pdf.

1 Altitude

GNSS throughout.
Check presence and range.

2 Fixes

Remove impossible positions;
invalidate bad altitudes.

3 Airborne interval

Trim ground phases;
split witnessed ground stops.

4 Flight selection

Duration ≥ 40 min
Path ≥ 20 km; altitude range
 ≥ 600 m
Integrity and reach checks

5 Coordinates

Local east and north.
Keep GNSS altitude.
One origin per flight.

6 Time grid

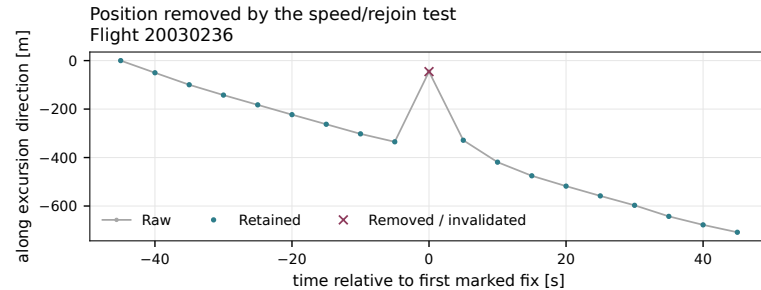
Fill short gaps; split long gaps.
Flag repairs; smooth
and differentiate.

A local outlier flag alone never deletes a fix; impossible steps drive removal

Cleaning and filtering: the full procedure

This slide compresses the entire preparation pipeline. GNSS altitude is used end to end; eligible, locally complete barometry is only corroborating evidence for specific cleaning decisions. Horizontal deletion uses physical reachability and a successful return to the trend; the Hampel identifier flags local disagreement but does not gate deletion. Timestamp defects, repeated-position runs and altitude defects have separate rules. Sustained movement defines the outer airborne interval, and sufficiently witnessed interior ground intervals are excised. Flight-level thresholds restrict the population: recorded duration at least 40 minutes, path at least 20 km, altitude range at least 600 m, elapsed span at most 16 hours, plus integrity and reach checks. Coordinates are projected locally, then resampled on the flight's native grid. The gap cap is $\min(10 \text{ times the native cadence}, 20 \text{ seconds})$, including gaps between finite altitudes. A Savitzky–Golay step supplies smoothed positions and derivatives. Source: thesis Sections 2.7–2.8 and the two cleaning ADRs.

Retain supported motion, record every repair



Real example:
remove an impossible excursion when the track rejoins.

Long unsupported intervals become boundaries.

The real example shows a horizontal excursion rejected by the speed/rejoin test, not merely by a local outlier flag. The point of cleaning is to retain supported motion while exposing where information has been removed or reconstructed. It does not certify that every remaining fix is error-free. The full cascade retains 156,406 paraglider flights out of 186,052, and 6,094 hang-glider flights out of 6,716. The percentages on the slide refer to retained flights, whereas 0.58 and 0.30 percent refer to reconstructed grid points within the retained output. These are different denominators. Reconstruction flags are kept, long gaps cause segment boundaries, and transport increments never cross a boundary. These counts describe the cleaned archive; further temporal-resolution and duration restrictions define the samples used below. Sources: generated/cleaning_real_examples.pdf and pipeline_census.tex, thesis Sections 2.7–2.8.

156,406 84.1% retained

Paragliders

6,094 90.7% retained

Hang gliders

Paraglider grid reconstruction: 0.58%

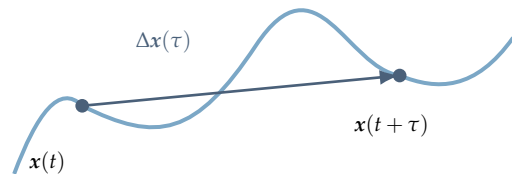
Hang glider: 0.30%

Measuring horizontal transport

Average within each flight,
then give each flight equal weight.

$$m_f(\tau) = \frac{1}{n_f} \sum_i |\Delta x_{fi}|^2$$

$$M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$$



Displacement is the straight-line change in position,
not distance travelled.

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusion} \quad H = 1 \text{ ballistic motion}$$

For each lag, form horizontal displacements between positions within the same retained segment. The displacement is a chord, not the path length. First average squared displacements over every admissible origin in a flight, pooling its selected segments. Then average over flights with equal weight. This prevents long flights from dominating through their number of origins and prevents heavily fragmented flights from gaining weight through their number of segments. For the main analysis, native cadences above ten seconds are excluded and each selected segment is sampled on a common ten-second grid. The 33 physical lags range from ten to ten thousand seconds. A log-log slope is written as $2H$; H is an effective growth exponent here, not an established self-similarity parameter of a stochastic process. The standard diffusive and ballistic values are references. Source: thesis Section 3.1.1.

Keep the same flights at every lag

7

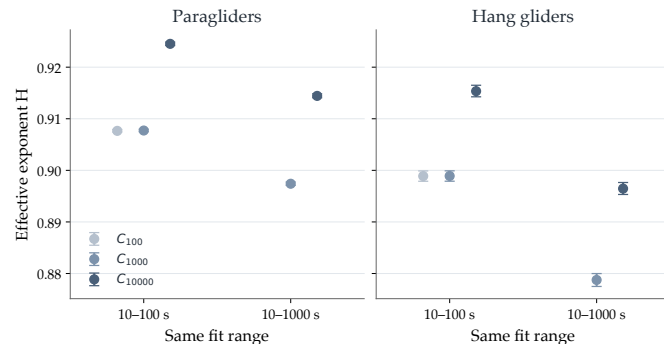
One cohort for 10–10,000 s

$$\tau_{\max} \leq 0.8T_s$$

Selected segments last
at least 12,500 s (3 h 28 min).

46,273 paragliders

2,326 hang gliders

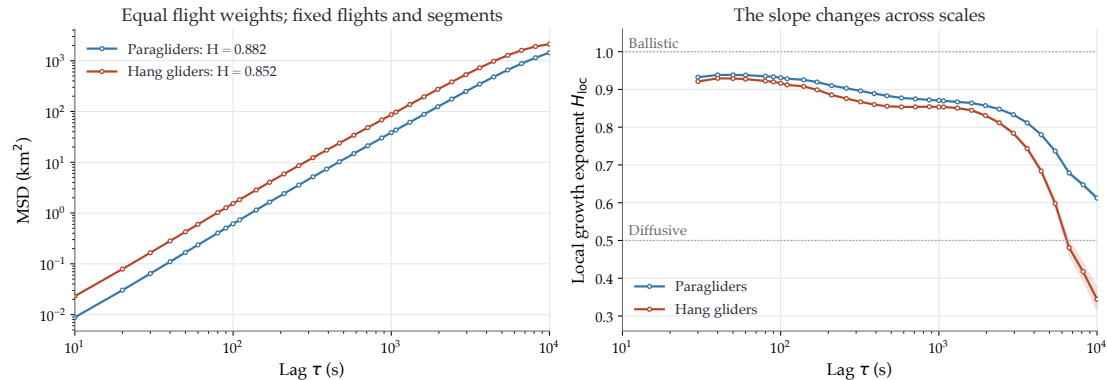


Selecting long segments raises H by about 2% on matched lag ranges

Keep the same flights at every lag

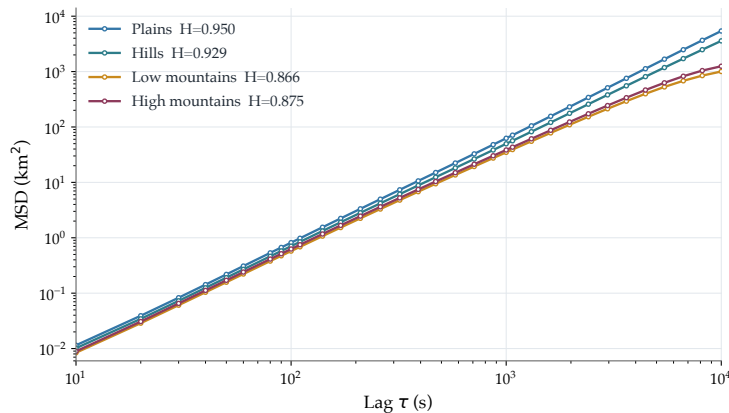
The available population loses short flights as lag increases. We therefore select a fixed set of flights and segments once, before computing the main curves. Requiring the maximum lag to be at most 80 percent of segment duration leaves at least one fifth of each selected segment available for origins at the longest lag. For ten thousand seconds, that means at least 12,500 seconds on the common grid and at least 251 origins at the maximum lag. Shorter segments of an otherwise eligible flight are excluded at every lag. The main cohort contains 46,273 paragliders and 2,326 hang gliders. The figure compares nested cohorts over identical fit intervals, using paired bootstrap draws. The main cohort increases H by 0.0164–0.0177, about 1.8–2.0 percent. We accept this small, measured duration-selection shift to keep the population constant across three decades of lag. Origins still vary with lag. Source: thesis Section 3.1.3 and ch3_transport_report.json.

Fast growth persists, but one exponent is a summary



Fixed population: $H_{\text{para}} = 0.882$ $H_{\text{hang}} = 0.852$ over 10–10,000 s

The fixed-population curves still show growth between the diffusive and ballistic references when summarised over ten to ten thousand seconds. The fitted exponents are 0.882 [0.881,0.883] for paragliders and 0.852 [0.849,0.855] for hang gliders. The local slope changes with lag, so the global fit compresses a curved relationship into a single effective number. Uncertainty is evaluated with one thousand resamples of take-off-site and calendar-day groups, retaining dependence among flights in a group. The same draws are reused over lags and, below, over conditional groups. The bands are pointwise 90 percent intervals, not a simultaneous band for the entire curve. A site-day grouping also does not guarantee independence between all groups; shared weather can remain. These results constrain a model but do not identify a unique stochastic process. Source: thesis Section 3.1 and the fixed-population arrays and fits in `ch3_transport_report.json`.



Initial GNSS altitude

Plains: < 300 m

Hills: $300\text{--}800$ m

Low mountains:
 $800\text{--}1,500$ m

High mountains:
 $\geq 1,500$ m

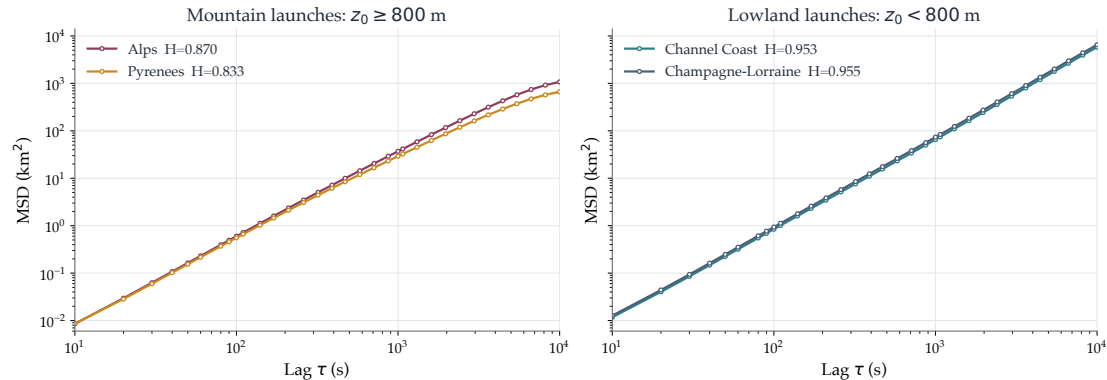
Above sea level, at the first
cleaned fix.

Pooled lowland curves grow faster; circuit composition also changes with altitude

Launch-altitude groups have different MSDs

The four groups partition the fixed paraglider cohort by GNSS altitude at the first cleaned fix. Altitude is measured above sea level. The conventional labels are Plains, Hills, Low mountains and High mountains; they do not measure local slope or terrain along the route. Their sizes are 1,844, 2,908, 22,097 and 19,424. The effective exponents are approximately 0.950, 0.929, 0.866 and 0.875. The high-mountain exponent is slightly larger than the low-mountain one, so even the pooled ordering is not monotonic in altitude. Circuit composition is very different: the open-route shares are 91.1 and 71.1 percent in the two lowland groups, compared with 33.5 and 35.0 percent in the mountain groups. Consequently the pooled difference is an association, not an isolated effect of orography. We resolve the circuit dependence on slide 13. Source: thesis Section 3.2.1 and ch3_conditional.json.

Region matters within a broad launch environment



Alps–Pyrenees: $\Delta H = 0.037$ Lowland pair: $\Delta H = -0.002$, interval includes zero

This comparison holds the broad altitude setting fixed within geographic boxes. The Alpine and Pyrenean groups retain only launches at or above 800 metres; the two lowland groups retain only launches below 800 metres. There are 32,790 Alpine flights, 2,257 Pyrenean flights, 1,068 Channel Coast flights and 798 Champagne-Lorraine flights. Alpine H is 0.870 versus 0.833 in the Pyrenees; their paired difference is 0.037 [0.033,0.041]. The lowland exponents are 0.953 and 0.955, with difference -0.002 [-0.005,0.001]. A similar exponent does not imply identical motion: displacement amplitudes and curve shapes can differ. These selections do not match weather, route composition, exact altitude, site or equipment. The regional contrast is descriptive and does not identify the effect of mountain geometry or thermal triggers. Source: thesis Section 3.2.1; ch3_conditional.json.

Viewer: intersect climb-labelled trajectories with horizontal planes.

Within each cell: same dates, segmentation and map bounds at both heights.

Mountains

$z_{AGL} = z_1$

Mountain cell
lower plane

Viewer screenshot

$z_{AGL} = z_2$

Mountain cell
higher plane

Viewer screenshot

Plains

$z_{AGL} = z_1$

Plains cell
lower plane

Viewer screenshot

$z_{AGL} = z_2$

Plains cell
higher plane

Viewer screenshot

Working observation: mountain patterns shift less with height than lowland patterns.

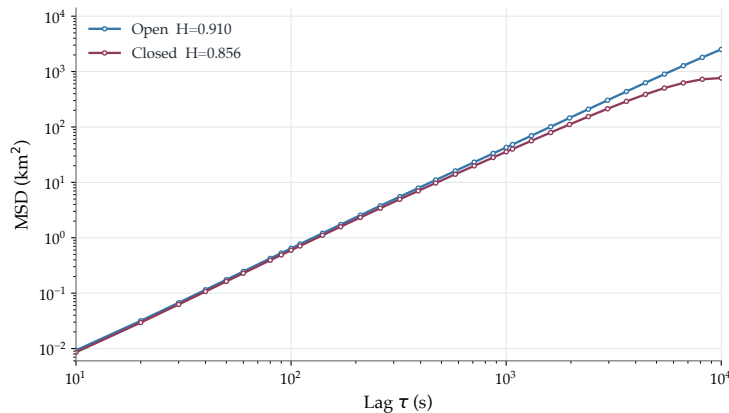
Possible explanations: stronger uplift or weaker wind; neither is measured here.

Dots are climb crossings, not thermal centres. z_{AGL} is relative to a fixed reference for each cell.

Thermal patterns across height

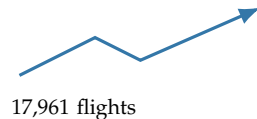
The viewer lets us inspect the spatial pattern of observed climb crossings at different horizontal planes. We compare the same mountain cell at two heights and the same plains cell at those heights, keeping the selected dates, segmentation method and map bounds fixed within each pair. The qualitative observation is that the mountain pattern appears more vertically aligned, whereas the lowland pattern shifts more as the plane rises. This concerns horizontal position as a function of height, not stationarity in time and not a change in the vertical coordinate of a fixed plane. Stronger uplift could shorten the time available for horizontal advection over a given height increment; weaker horizontal wind could also reduce that displacement. Terrain-anchored lift is another possible contributor. These are hypotheses, not causes established by these views. The dots are observed climb crossings, possibly several per thermal, not identified thermal centres or independent thermals. Different flights can support the two heights. In this viewer, AGL is height above a fixed cell reference derived from starting altitudes, not pointwise height above a terrain model. Source: docs/guide/thermal-planes.md; qualitative interpretation proposed by the presenter.

Open routes have larger displacement growth

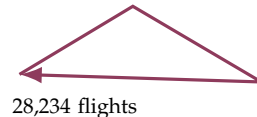


$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] \quad (90\% \text{ paired interval})$$

Open circuits

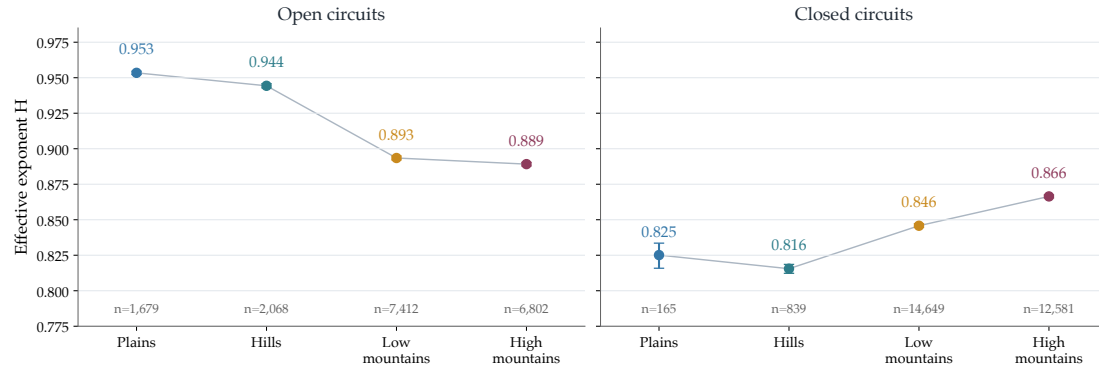


Closed circuits



The catalogue describes the intended circuit, not a guarantee that the recorded trimmed trajectory returns to its starting point. Closed routes include triangles and out-and-return flights. Pooling launch altitudes gives $H=0.910$ for open routes and 0.856 for closed routes. Their paired difference is clearly positive under the adopted bootstrap. Turning back can reduce net displacement over intervals spanning a turn. However, lag is averaged over origins throughout a segment: there is no universal onset at half or one third of flight duration, and ten thousand seconds is an analysis limit, not a shared total duration. The route schematics illustrate geometry only. The pooled comparison still mixes launch environments and other conditions, so its magnitude cannot rank the causal importance of route choice against terrain. Sources: thesis Section 3.2.2, ch3_conditional.json.

The altitude ordering reverses for closed routes

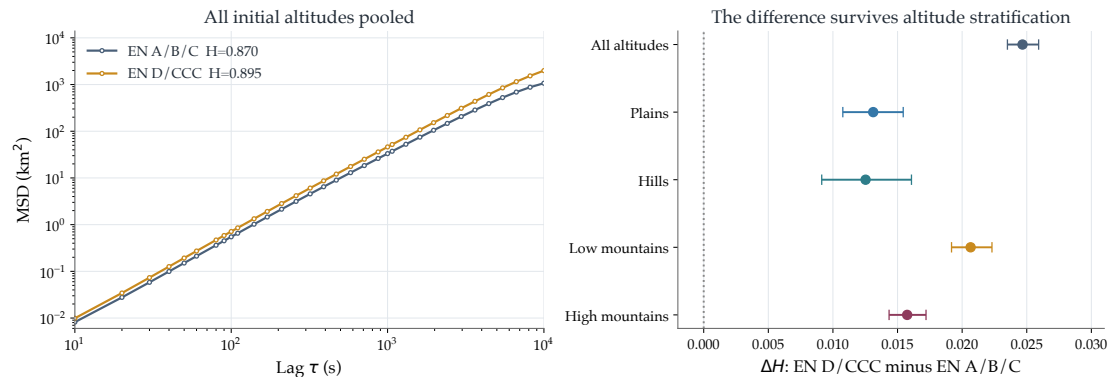


Open: lowland H is higher

Closed: mountain H is higher

This is the key joint comparison. The plotted points are fitted exponents from the same four altitude classes, separated into open and closed circuits. For open routes, the lowland-minus-mountain contrasts are positive, between 0.051 and 0.064. For closed routes, they are negative, with magnitudes between 0.021 and 0.051. All eight paired intervals exclude zero. Open routes also have larger H than closed routes within each altitude class. The smallest cell is 165 closed flights in Plains, so its uncertainty is visibly larger. The observed altitude association changes sign with circuit type; a universal claim that mountain launch environments suppress transport growth is therefore not supported. Possible mechanisms include the organisation of lift, detours and differences in circuit size or turn times, but these are hypotheses. The figure establishes conditional orderings, not their causal explanation. Source: thesis Section 3.2.2; ch3_conditional.json and ch3_transport_task_table.tex.

The equipment contrast survives altitude stratification



Higher H for EN D/CCC in every altitude class; largest gap in Low mountains

The two equipment groups contain 27,609 flights on EN A/B/C wings and 16,497 on EN D/CCC wings; 2,167 flights are excluded from this comparison. The thesis calls them beginners and experts as an operational proxy. Individual experience is not observed: an experienced pilot can fly a lower-class wing, and the same pilot can contribute to different groups on different flights. The pooled exponent difference is 0.025 [0.024,0.026]. Within altitude classes the differences are 0.013, 0.013, 0.021 and 0.016, all with paired intervals excluding zero. The Low mountains contrast exceeds the others in the paired interaction tests. A constant speed multiplier alone changes MSD amplitude, not its log-log slope, so the difference concerns the lag dependence too. It still mixes wing performance, pilots, routes and conditions. Source: thesis Section 3.2.3, ch3_conditional.json and ch3_conditional_equipment_table.tex.

01

Rapid horizontal spreading in both disciplines,
with an effective exponent that changes across scales.

02

The altitude association depends on the declared circuit:
the lowland–mountain ordering reverses for closed routes.

03

Equipment-group differences remain within altitude classes;
their physical and behavioural contributions are still mixed.

Next: explain global transport through flight behaviour and physical models

What the measurements establish

The work so far delivers a traceable dataset and a set of empirical constraints for transport modelling. Both disciplines show rapid displacement growth, but a single effective exponent is not the whole story. Keeping flights and segments fixed separates lag dependence from changing membership. Among long paraglider flights, launch environment, region, route and equipment all carry structure. Most notably, the altitude ordering reverses between open and closed routes, and the equipment contrast persists within altitude classes. These are observed associations, not isolated causal effects. The next stage is to connect the global results to flight behaviour and test physical stochastic models against several observables under the same measurement protocol. The thermal-plane view adds a qualitative observation about vertical alignment. It does not establish a quantitative thermal-displacement law or identify its mechanism. Sources: current thesis introduction and Sections 3.1–3.2; thermal-plane viewer documentation.